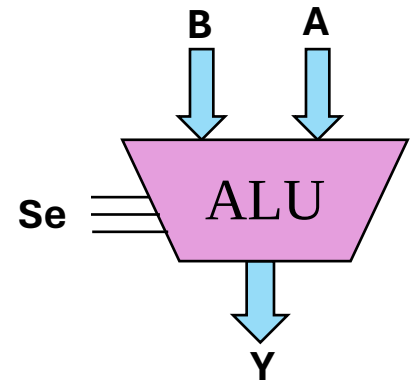


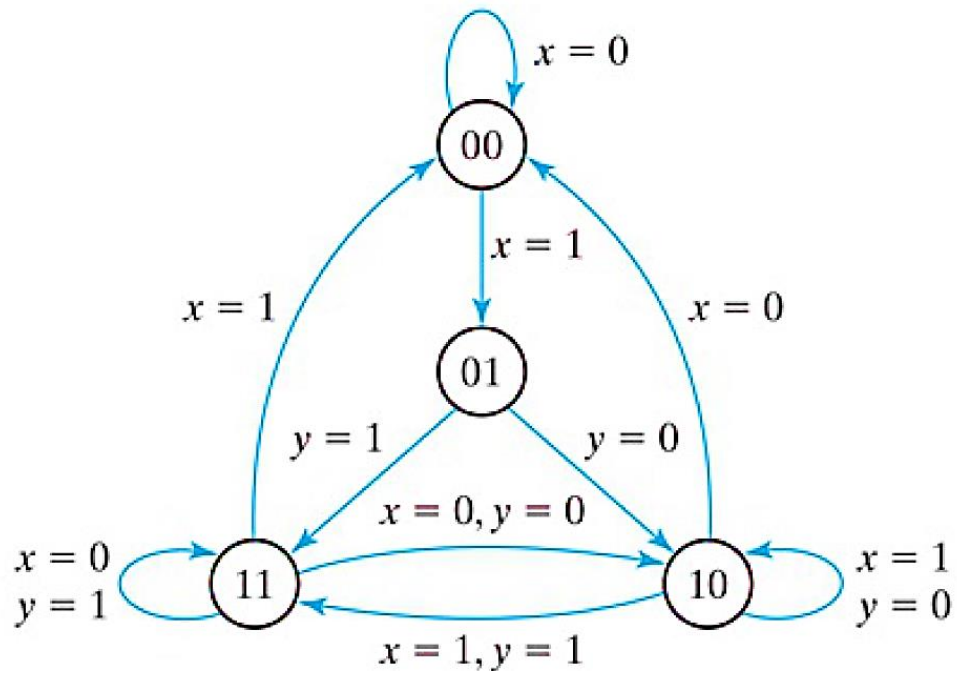
Note: in the following questions, the behavioral description that you write can be in VHDL, Verilog, SystemVerilog, C, Python, or MATLAB.

1. Write a behavioral description of an eight-bit arithmetic-logic unit (ALU). The circuit has a three-bit select line (Sel), 16-bit inputs (A and B), an eight-bit output (Y), and performs the arithmetic and logic operations listed below.

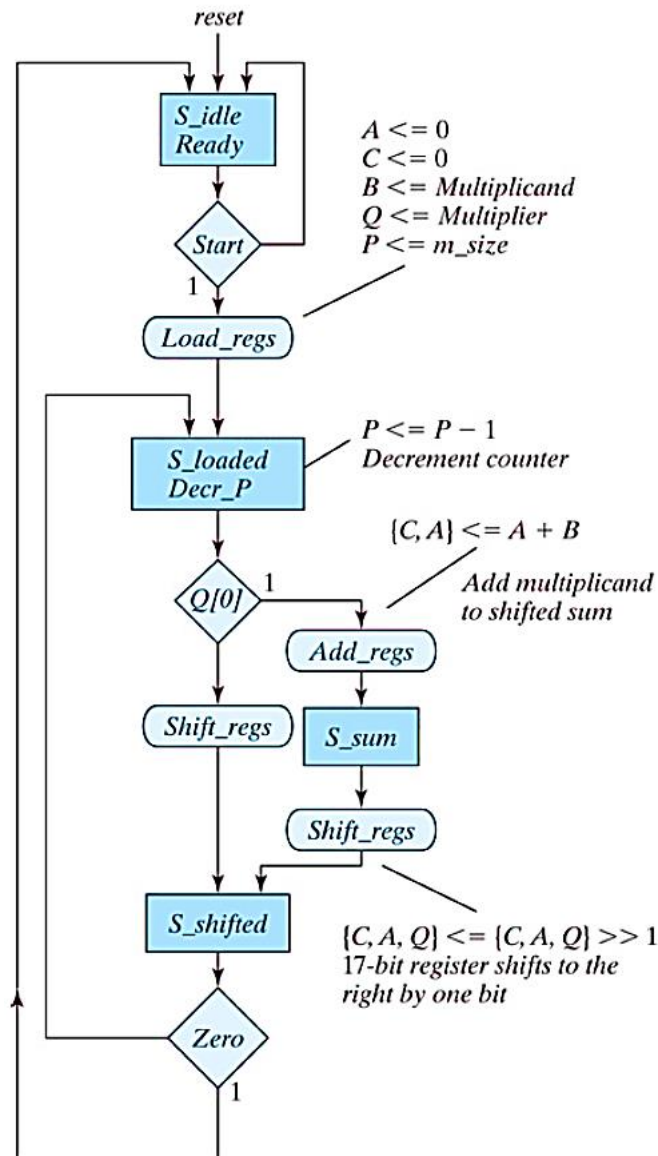
Sel	Operation
000	Reset Y to all 0's
001	Bitwise AND
010	Bitwise OR
011	Bitwise exclusive-OR
100	Bitwise complement
101	SUB (Subtract)
110	ADD (Assume A and B are unsigned)
111	Set Y to all 1's



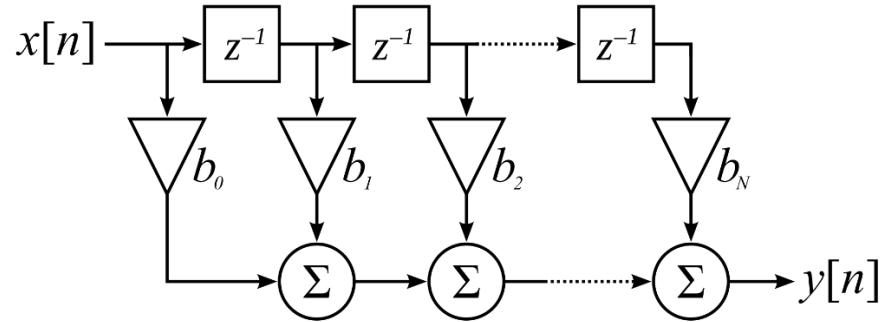
2. Write a behavioral description model of the state diagram shown below.



3. Figure below shows an Algorithmic State Machine (ASM) with Datapath (ASMD) chart for a sequential binary multiplier. Write a behavioral description model of the system.



4. Consider an N-tap FIR filter, as shown below.



where, $x[n]$ and $y[n]$ are the input and output arrays, Z^{-1} represents a memory (delay) unit, which can be implemented using a register, and b_i are filter coefficients. Assuming 16-bit fixed-point numbers.

- Write a behavioral description for this filter.
- Design scheduling diagram of the hardware implementation of this filter without pipelining.
- Design scheduling diagram of the hardware implementation of this filter with pipelining.
- Compare these designs in terms of throughput and hardware costs.

Q1: Given the following behavioral HDL snippet, translate it into a Control Data Flow Graph (CDFG):

```
1  process(clk, reset)
2  begin
3      if rising_edge(clk) then
4          if reset = '1' then
5              counter <= (others => '0');
6          else
7              counter <= counter + 1;
8          end if;
9      end if;
10 end process;
```

Q2. You are given a basic sequence of operations required for a certain computation task in a digital signal processor design. These operations must be performed under the constraints of a specific execution order due to data dependencies.

Given Operations and Dependencies:

Operations need to be performed: A, B, C, D, E, F, G, H

- **Execution Times:**

Operation	A	B	C	D	E	F	G	H
Latency (in clock cycles)	1	2	1	2	3	1	1	2

- **Dependencies:**

- A needs to be done before B and C.
- B needs to be done before D.
- C needs to be done before D and E.
- D needs to be done before F.
- Both E and F need to be done before G.
- G needs to be done before H.

Task 1: ASAP Scheduling

- Schedule the given operations using the As Soon As Possible (ASAP) scheduling algorithm. Assume each operation starts as soon as its prerequisites are completed, and there are unlimited resources.

Task 2: ALAP Scheduling

- Using the total execution time found in Task 1 as the deadline, perform As Late As Possible (ALAP) scheduling for the same set of operations. Consider the same dependencies and execution times.

Task 3: Mobility Calculation

- For each operation, calculate its mobility based on the ASAP and ALAP schedules. Recall that mobility is defined as the difference between the start times in the ALAP and ASAP schedules.

Expected Outputs:

Students are expected to produce:

1. **ASAP Schedule:** A chart or table indicating the start times of each operation in the ASAP schedule.
2. **ALAP Schedule:** A chart or table showing the start times for each operation in the ALAP schedule given the total execution time constraint from Task 1.
3. **Mobility for Each Operation:** A list or table with each operation and its corresponding mobility value.

Q3. Consider the Discrete Fourier Transform (DFT) equation for an input signal ($x[n]$) of length N :

$$X[k] = \sum_{n=0}^{N-1} x[n] \times e^{-j\frac{2\pi}{N}kn}$$

where:

- $X[k]$ represents the frequency domain representation at frequency index (k).
- $x[n]$ is the input signal in the time domain at time index (n).
- $e^{-j\frac{2\pi}{N}kn}$ is the complex exponential kernel used in DFT calculations (Note: This is a complex number).

Task:

1. Convert the given mathematical expression into a Control Data Flow Graph (CDFG).
2. Propose a design using standard functional units to execute the operations defined in the expression.

Expected Outputs:

Students should provide:

1. **CDFG Representation:**
 - A detailed graphical representation translating the complex expression into a CDFG. This should include the handling of the summation and product sequences, the calculation of the function ($X[k]$).
2. **Hardware Design Using Functional Units:**
 - A schematic or detailed description of how the operations identified in the CDFG map onto standard functional units such as multipliers, adders, and other necessary components.

Q4. Design a pipelined arithmetic unit for performing a Discrete Wavelet Transform (DWT) using the Haar wavelet transform. The DWT equation for the Haar wavelet transform can be represented as follows:

$$c_k = \frac{a_{2k} + a_{2k+1}}{\sqrt{2}}, \quad d_k = \frac{a_{2k} - a_{2k+1}}{\sqrt{2}}$$

where:

- (a_k) represents the input signal at index (k) .
- (c_k) and (d_k) are the approximation and detail coefficients at level (k) .

Task:

Design a **pipelined arithmetic** unit that efficiently computes the Haar wavelet transform using the DWT equation provided above.

Expected Outputs:

- Develop a detailed **DFG** and **Hardware Design Using Functional Units** illustrating the pipelined structure of the arithmetic unit for implementing the Haar wavelet transform using the DWT equation.
- Calculate the pipeline latency and overall **throughput** for processing the 8-point input signal through the DWT computation.
- Evaluate the hardware implementation **costs** and resource utilization associated with the pipelined DWT unit compared to a non-pipelined design.

